(Kissell and Malamut, 2006), and Institutional Investor's Guide to Algorithmic Trading, "Understanding the P&L Distribution of Trading Algorithms," (Kissell and Malamut, 2005). Additionally, Almgren and Lorenz analyzed real-time adaptive strategies in Institutional Investor's Algorithmic Trading III: Precision, Control, Execution, "Adaptive Arrival Price," (Spring 2007), and also in Journal of Trading's "Bayesian Adaptive Trading with a Daily Cycle," (Fall 2006).

We now expand the previous research findings and provide an appropriate algorithmic framework that incorporates both macro and micro decisions. Our focus is with regards to single stock trading algorithms and single stock algorithmic decisions.

In Chapter 9, Portfolio Algorithms, we provide an algorithmic decision making framework for portfolio and program trading.

EQUATIONS

The equations used to specify macro and micro trading goals are stated below. Since we are comparing execution prices to a benchmark price in \$/share units our transaction cost analysis will be expressed in \$/share units. For single stock execution (in the US) this is most consistent with how prices and costs are quoted by traders and investors. Additionally, our process will incorporate the trading rate strategy α but readers are encouraged to examine and experiment with this framework for the percentage of volume and trade schedule strategies. These transaction cost models were presented in Chapter 7, Advanced Algorithmic Forecasting Techniques.

Variables

$$I' = Instantaneous\ Impact\ in\ \$/Share$$

$$MI' = Market\ Impact\ in\ \$/Share$$

$$TR' = Timing\ Risk\ in\ \$/Share$$

$$PA' = Price\ Appreciation\ Cost\ in\ \$/Share$$

$$X = Order\ Shares$$

$$Y = Shares\ Traded\ (Completed)$$